

Astem: Baku Metro Heat Recovery

Transforming transit into clean energy.



ASTEM Core Team

Roles aligned for pilot execution

TEAM LEADER

Isa Ibadov

Strategy, partnerships, and investor relations.

TECHNICAL MEMBER

Shukurov Shukur

Prototype design and installation specifications.

FUNDER

Rufat Jabrayilli

Business model, financial projections, and grant search.

RESEARCH TEAM

**Ziya Aliyev • Shargiyya Qafarzade •
Aysel Sardarova**

Field measurements, technical models, and scientific basis.

Six-member execution team built across strategy,
engineering, research, and finance.



Trains generate massive heat.
High passenger density traps warmth.



Valuable energy lost to atmosphere.

Capturing Baku's Wasted Subterranean Energy

The Paradox

Baku Metro tunnels stay between 18°C–30°C year-round.

Currently, this energy is wasted—blown out through ventilation shafts while nearby buildings burn fossil fuels for heat.

The Concept

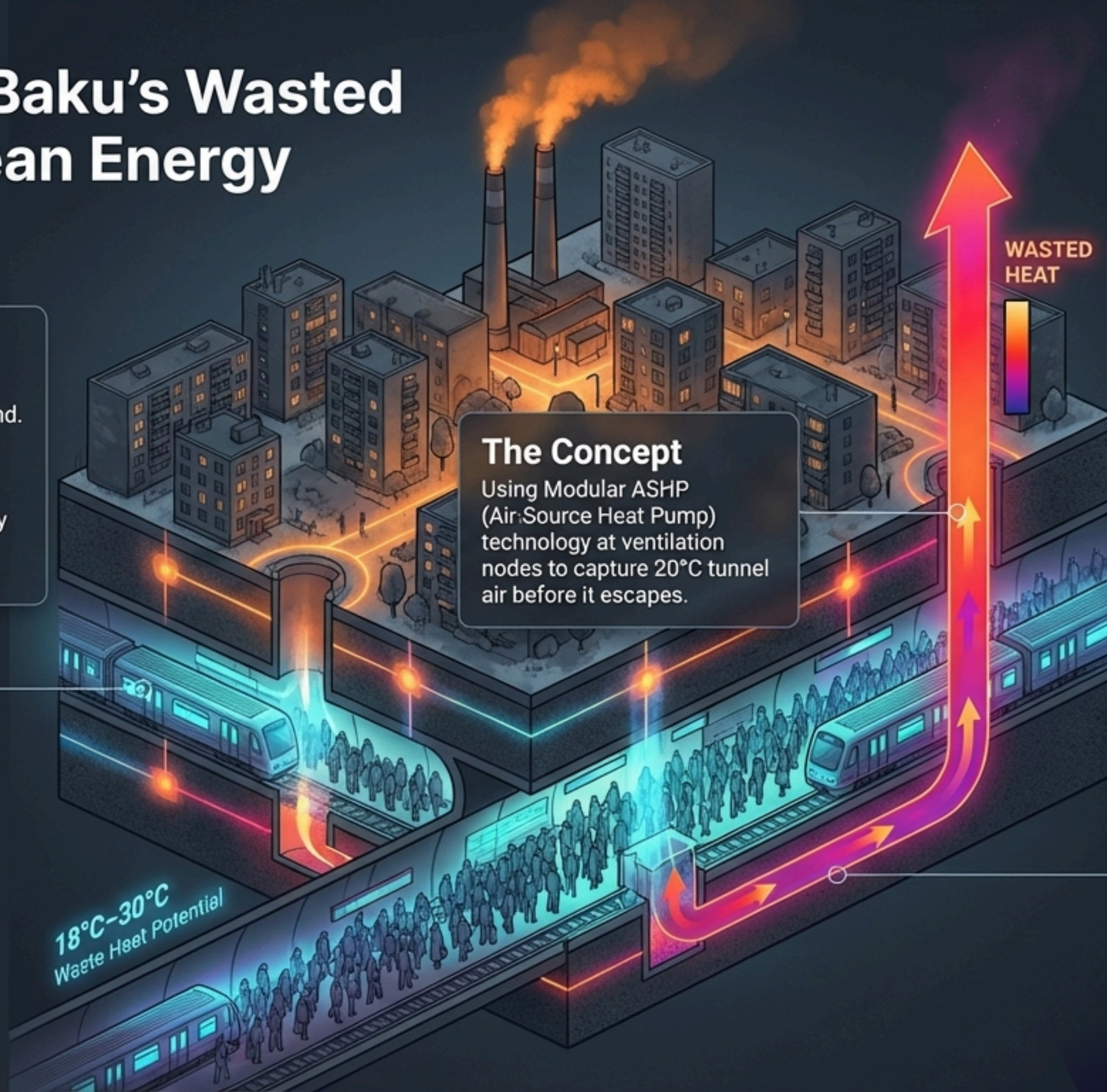
Using Modular ASHP (Air-Source Heat Pump) technology at ventilation nodes to capture 20°C tunnel air before it escapes.

The Opportunity

Transform 'waste heat' into a sustainable, 60°C–80°C high-temperature water supply for urban district heating.

18°C–30°C
Waste Heat Potential

WASTED
HEAT



Higher density equals higher efficiency.



3.7 million
passengers/station.

London

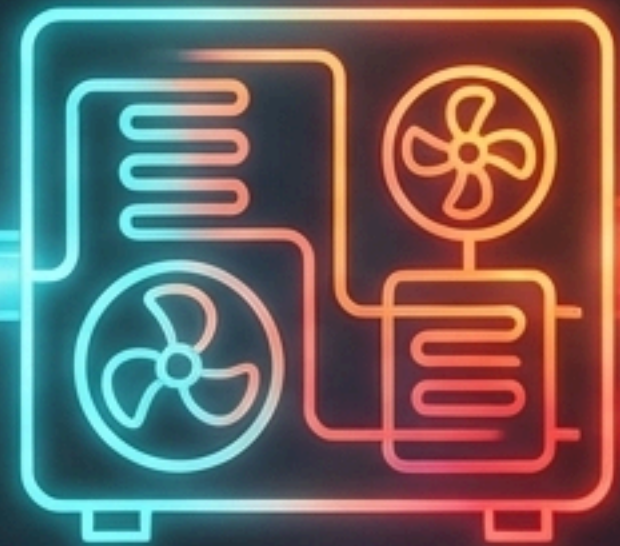


8.3 million
passengers/station.

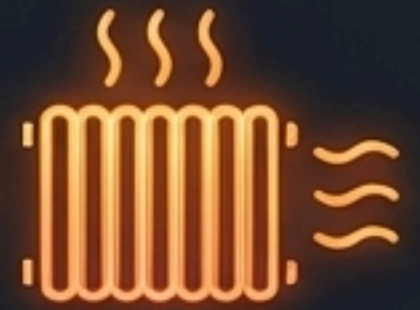
Baku



Capture 5–30°C
tunnel air



Upgrade via Air
Source Heat Pumps



Deliver 50°C
building heating



Supply 60°C
domestic hot water

- MAP LAYERS
- ☒

Waste Heat Potential (Volume)
- ☒

Metro Exits (Hotspots)
- ☒

Metro Network (Lines)

BAKU METRO IMPACT (2025)



2,153,000

Total Daily Pax



541,908 kW

Recoverable Heat



474,711.4

CO₂ Reduction (Tons)



237,356

Homes Heated



2,373,557 MWh

Energy Recovered



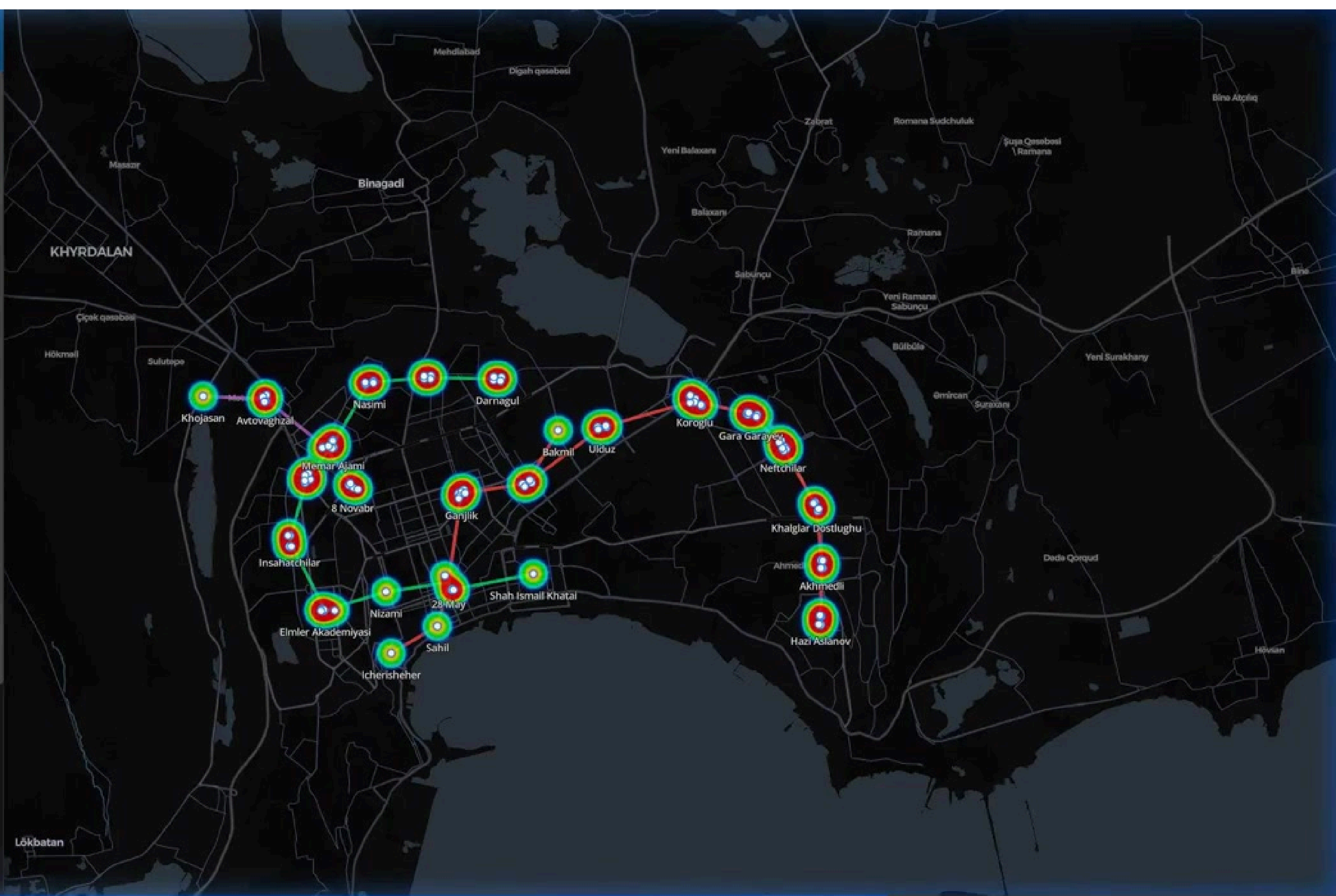
27

Measured Stations

LINE INFORMATION

- 

Red Line
- 13 Stations | 18.8 km

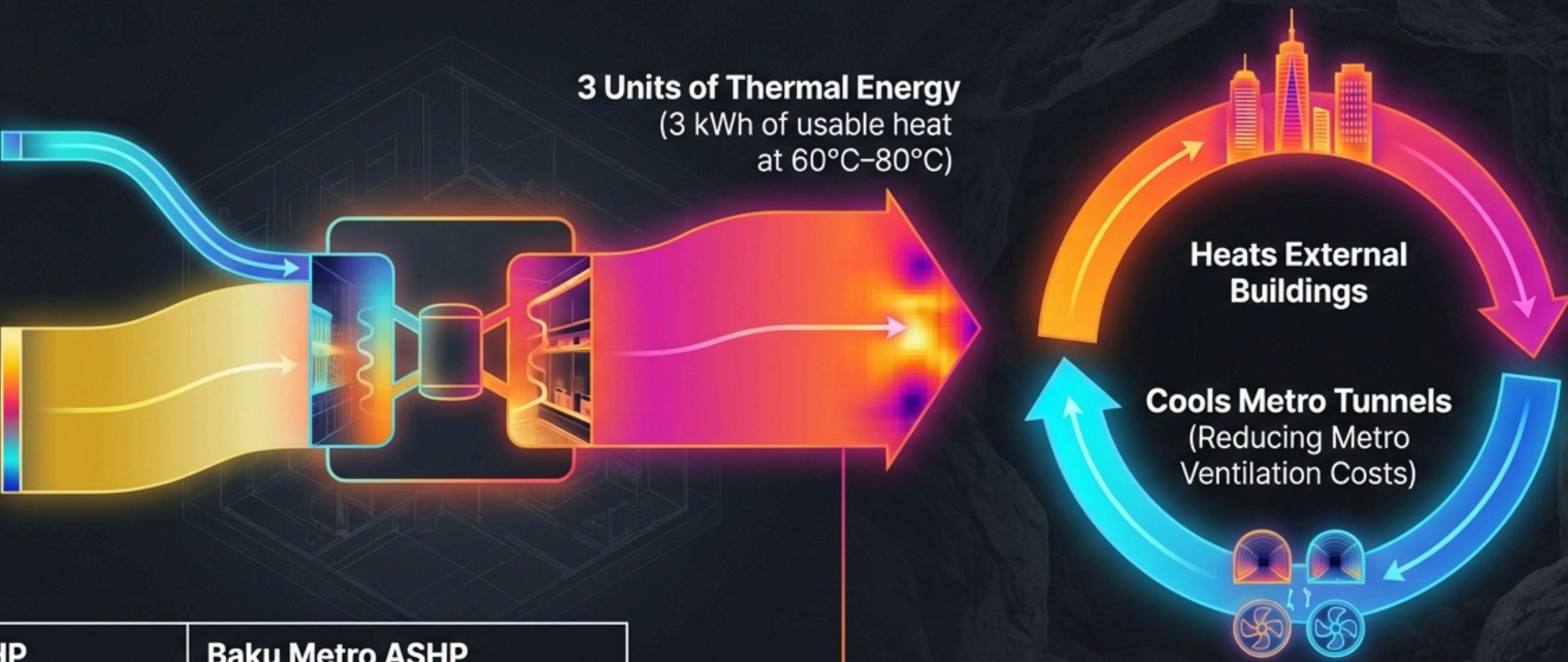


Heat Pump multiplier engine

1 Unit of Electricity
(1 kWh)

2 Units of Ambient Tunnel Heat
(20°C)

3 Units of Thermal Energy
(3 kWh of usable heat at 60°C–80°C)



Standard ASHP

Pulling from 0°C freezing winter outside air results in high electricity demand, low efficiency, and high cost.

Baku Metro ASHP

Pulling from 20°C stable tunnel air results in low electricity demand, high efficiency (COP 3.0), and high profitability.

Global Precedents: Based on the successful London Bunhill 2 project (heating 500+ homes) and Warsaw M2.

Summer Mode



Summer:
Cools stations,
heats water.



Winter Mode

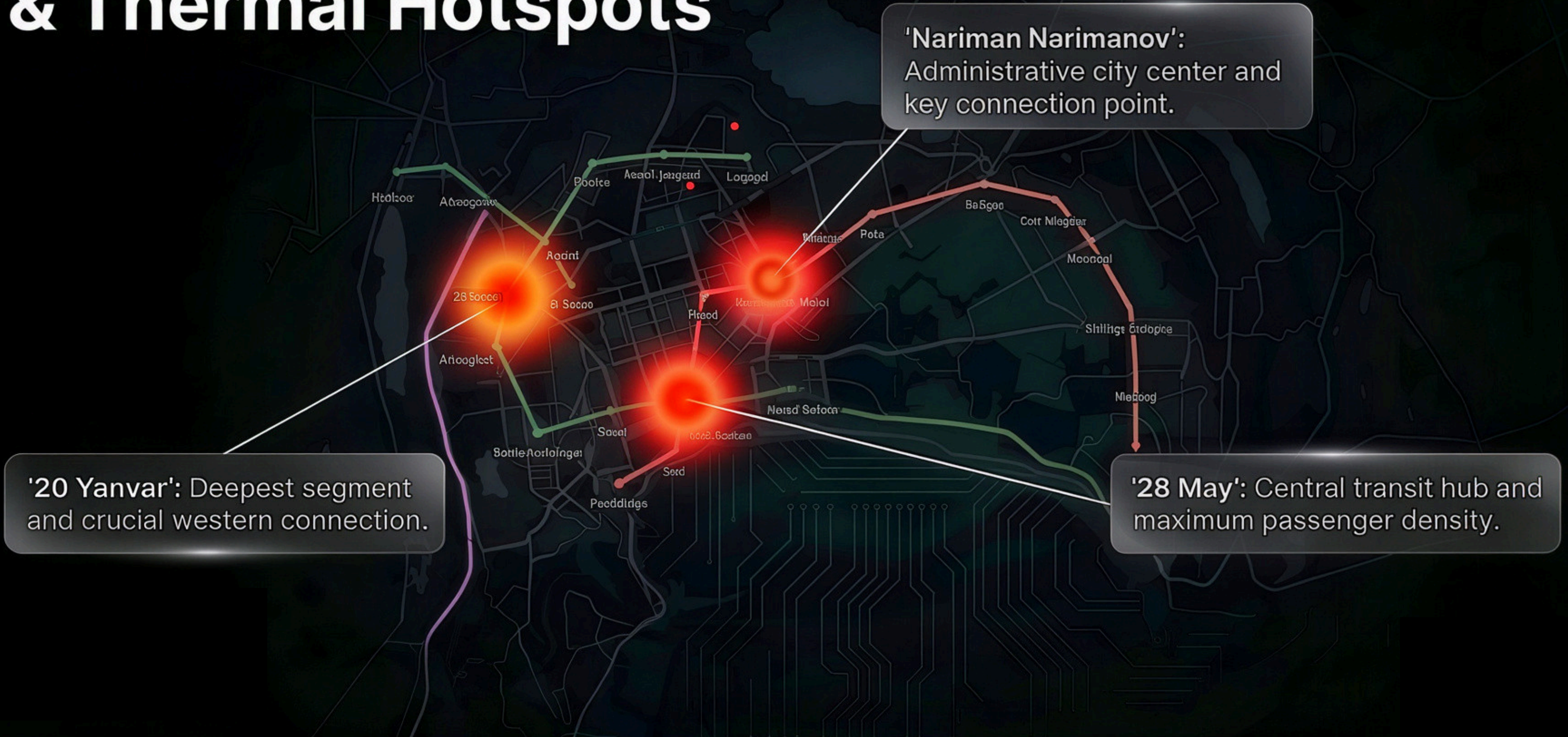


Winter:
Recovers heat
efficiently.



Prioritizes passenger comfort year-round.

Baku Metro Network & Thermal Hotspots



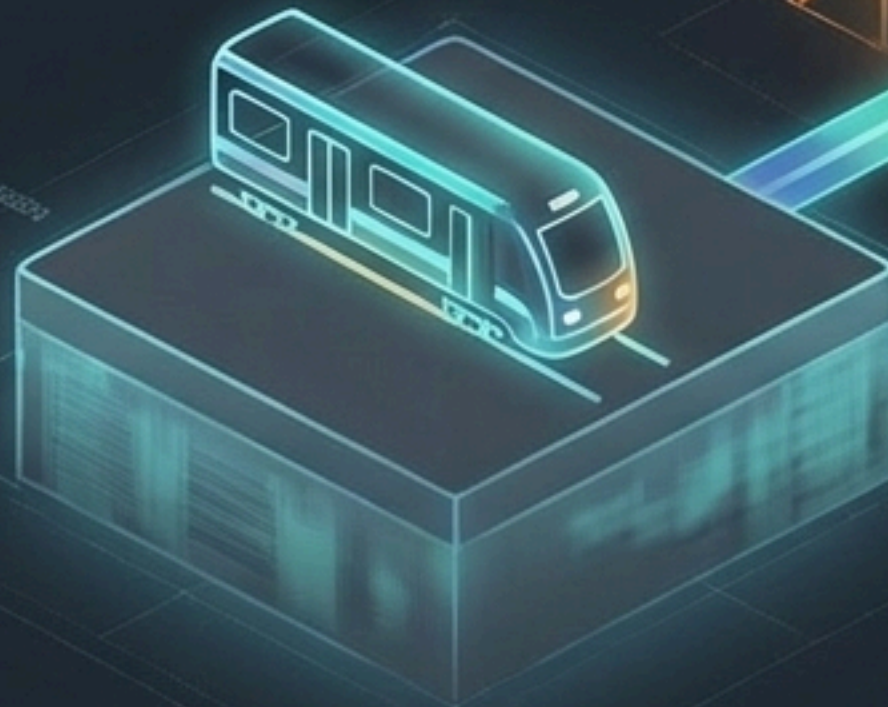
A Strategic Roadmap to Profitability

Phase 1: Internal Optimization

Client: Baku Metro Authority

Product: Cooling-as-a-Service

Value: Reduces tunnel temperatures and extends equipment lifespan without relying solely on the 82 energy-intensive ventilators currently in use.



Phase 2: External Commercialization

Client: Nearby Residential Complexes & Business Centers

Product: District Heat

Value: Selling upgraded, 60°C hot water directly to local stakeholders at a rate cheaper than traditional grid electricity or gas heating.

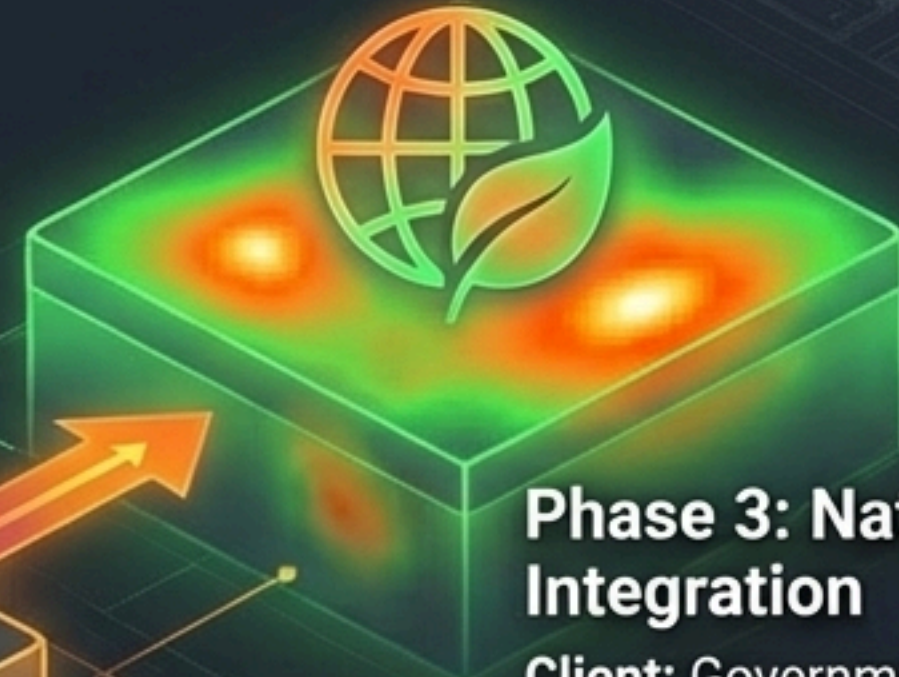


Phase 3: National Integration

Client: Government & Global Markets

Product: Zero-Emission Heating & Carbon Credits

Value: Direct alignment with Azerbaijan's 2030 Green Energy Goals, unlocking government subsidies and global carbon offset markets.



1.8 MW

Generate 1.8 MW thermal capacity.

11.9 GWh

Save 11.9 GWh annually.

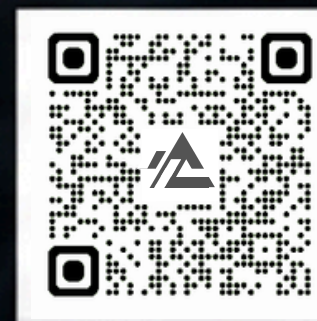
476,000 AZN

Unlock 476,000 AZN yearly savings.

7–10 Years

Achieve 7–10 year ROI.

**Fund the Phase 1 Pilot.
Validate the digital twin.
Scale across Baku Metro.
Transform urban energy forever.**



Join the
Astem Project